Shoaib Sadiq Salehmohamed

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Education

Indian Institute Of Technology–Madras — Chennai, Tamil Nadu
Bachelor of Science (BS) in Data Science and Applications — ***CGPA: 8.89/10*** — *May. 2024 – September 2028*

Keshav Memorial Institute of Technology — Hyderabad, Telangana
Bachelor of Technology in Computer Science and Engineering — ***CGPA: 8.87/10*** — *Aug 2024 – Aug 2028*

Technical Skills

Languages: Java, Python, C, SQL (PostgreSQL, MySQL), JavaScript, HTML/CSS, TypeScript
Frameworks: Next.js, Express.js, Vue.js, React, Node.js, Flask, FastAPI, ROS, ArduPilot
Developer Tools: Git, Docker, Google Cloud Platform, Azure, AWS, Linux (Ubuntu, Arch), Redis, Celery, WebSockets
Libraries: Pandas, NumPy, Transformers, Matplotlib, PyTorch, OpenCV

Problem Solving & Core CS

Solved **200+ LeetCode problems** primarily in **Java**; strong foundation in DSA, OOP, OS, DBMS, and computer networks.

Projects

TattvaDrishti | *Python, PyTorch, FastAPI, Next.js* — **SIH 2025 Winner – PS 25161**

- Built an end-to-end AI-driven cyber intelligence platform for detecting **LLM-powered misinformation** and coordinated influence operations using FastAPI.
- Fine-tuned DeBERTa-v3-Large models achieving **95% AI-vs-human accuracy** and **98% model-attribution accuracy** on a **2.7M+ sample corpus**.
- Developed graph intelligence, explainable risk scoring using **10+ signals**, real-time analyst dashboards via SSE, and a **blockchain-backed audit trail**.

Rakshak — **Autonomous Surveillance Rover** | *Python, Jetson Nano, LiDAR, GPS, Pixhawk*

- Led development of an autonomous surveillance rover integrating **LiDAR, GPS, IMU, and vision sensors** for navigation, obstacle avoidance, and perimeter monitoring.
- Implemented a **hybrid control architecture** where Pixhawk handled mission execution while Jetson Nano performed perception, decision-making, and safety overrides.
- Engineered **LiDAR-based collision avoidance**, real-time video streaming, telemetry, and on-device face recognition using WebSockets and OpenCV.

LingualLink — **Real-Time Translation Chat** | *FastAPI, Next.js, Redis, PostgreSQL, MongoDB*

- Built a real-time multilingual chat system with language detection, translation, and low-latency cross-language messaging.
- Used **Redis queues/PubSub** and batched GPU inference, reaching **∼20 ms/msg** latency and **∼48 translations/sec**.

Experience

Cyber Security Intern — Jun 2026 – Present
AICTE Cyber Security Cell — *Remote / India*

∗ Continuing development of **TattvaDrishti** into a real-world cyber intelligence application for threat analysis, misinformation detection, source verification, and trustworthy digital information systems.

Research Intern – LLM Lens — Feb 2026 – Present
AI Research Intern – LLM Reliability & Hallucination Detection — *Remote*

∗ Built hallucination detection systems across **15K+ samples(800+ GB)** using internal transformer representations, hidden-state probes, MLP/cross-layer/hierarchical architectures, and PyTorch/Hugging Face evaluation pipelines.

Open Source Contributor — Jul 2024 – Oct 2024
GirlScript Summer of Code (GSSoC) & Hacktoberfest — *Remote*

∗ Ranked in the **top 8% contributors** (**185 / 2613**) in GSSoC and recognized as a **Hacktoberfest Super Contributor** for open-source contributions.

Achievements & Publications

Smart India Hackathon 2025 Winner – PS 25161, AICTE Cyber Security Cell.
Published Indian Patent Application for Rakshak autonomous surveillance rover – **202541112724 A**.
arXiv:2604.06277 – Proposed hallucination detection using internal transformer representations with MLP, cross-layer, and hierarchical models.
arXiv:2606.07528 – Co-authored BEACON, a behavioral entropy based cross-model hallucination detection framework achieving **0.8123 AUROC**.